STUDYNEST | CLASS 11 PHYSICS

MARKING SCHEME AND MODEL ANSWERS

Chapter 3: Motion in a Plane

Maximum Marks: 40 | Suggested time: 60 minutes

Use after attempting the paper. Accept equivalent correct methods.

1 mark: Correct fact, formula, unit or direction.
2 marks: Principle/formula (1); explanation, working or result (1).
3 marks: Relation (1); working/reasoning (1); final result and unit (1).
4 marks: Principle (1); setup/equation (1); working (1); result/interpretation (1).
Mark splits are a guide; award partial credit for valid steps.

Vector.

Acostheta.

Zero.

Inward.

5.

v=romega.

45 deg .

Zero.

A_x=5 N; A=8.66 N.

Zero.

24 m/s^2.

2 s.

10.

4pi rad/s.

T=2 s; H=5 m.

10 N at 36.9 deg north of east.

4 m/s^2 inward.

3 s; 60 m.

2 s; 5 m; 34.6 m.

40 N inward; centripetal force does zero work.